

Data Science

CSE558

Recap

- Test (Correlation, Compare variance b/w two sets, goodness of fit, independence between two attributes)
- Principle of Inclusion & Exclusion

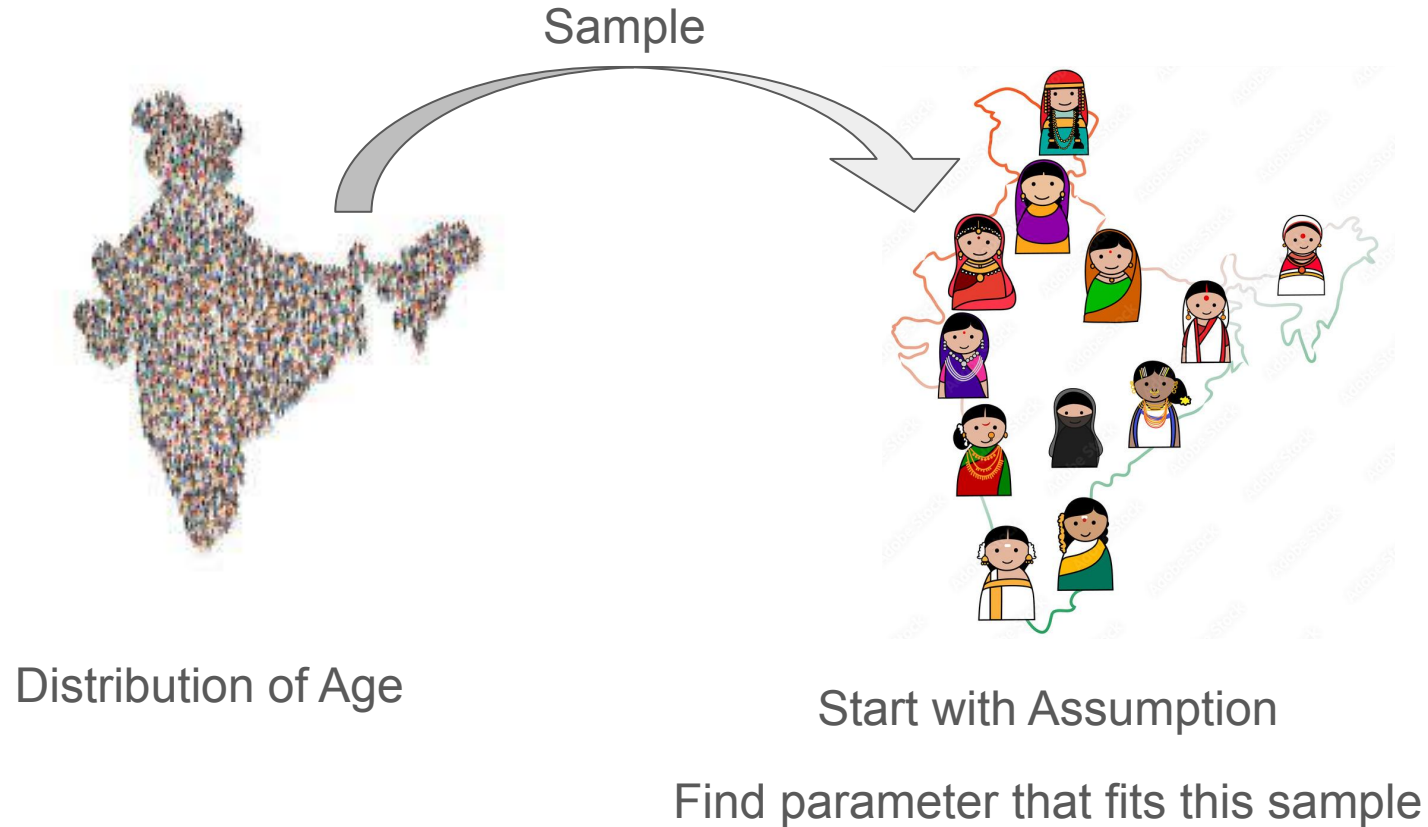
$$\mathbf{P} \left(\bigcup_{i=1}^n \mathbf{A}_i \right) \leq \sum_{i=1}^n \mathbf{P}(\mathbf{A}_i)$$

$$\mathbf{P} \left(\bigcup_{i=1}^n \mathbf{A}_i \right) \geq \sum_{i=1}^n \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i, j \leq n} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j)$$

Overview

- Likelihood Estimation
- Quiz - 1

Maximum Likelihood Estimation



Maximum Likelihood Estimation

Let there are random samples $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$, whose distribution depends on some unknown parameter Θ .

- Bernoulli (p)
- Binomial (p)
- Geometric (p)
- Normal (μ, σ)

The goal is to find the unknown parameter Θ that maximizes the likelihood of the samples $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$.

Maximum Likelihood Estimation (MLE)

Let there are random samples $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$, whose distribution depends on some unknown parameter Θ .

$$\mathbf{P}(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \mid \theta) = \prod_{i=1}^n \mathbf{P}(\mathbf{X}_i \mid \theta)$$

$$\frac{d\mathbf{P}(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \mid \theta)}{d\theta} = 0 \qquad \frac{d^2\mathbf{P}(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \mid \theta)}{d\theta^2} < 0$$

$$\log(\mathbf{P}(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \mid \theta)) = \sum_{i=1}^n \log(\mathbf{P}(\mathbf{X}_i \mid \theta))$$

Example

Consider a biased coin which has either $\frac{1}{3}$, $\frac{1}{2}$ or $\frac{2}{3}$ probability of flipping heads. Suppose we toss this coin for 10 times and observed 4 heads. Which probability distribution is most likely be true.

We assume the event follows binomial distribution. So for $n = 10$ and probability p .

$$P(\text{\#head} == 4) \text{ is } \binom{10}{4} p^4 (1 - p)^6$$

$$p = \frac{1}{3}, P(\text{\#heads} == 4) = 0.228$$

$$p = \frac{1}{2}, P(\text{\#heads} == 4) = 0.205$$

$$p = \frac{2}{3}, P(\text{\#heads} == 4) = 0.057$$

Example

Suppose we tossed a biased coin for 10 times and observed 4 heads. What is the maximum likelihood.

Let the unknown parameter be p . Then $P(\text{\#heads} = 4)$ is $\binom{10}{4} p^4 (1 - p)^6$

$$\frac{\binom{10}{4} p^4 (1 - p)^6}{dp} = \binom{10}{4} (4p^3 (1 - p)^6 - 6p^4 (1 - p)^5)$$

At, $p = 0$, $\frac{2}{5}$, & 1 the above expression is 0 .

$p = \frac{2}{5}$.

MLE of Bernoulli Distribution

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are random samples that corresponds to your n independent experiments. Let $\mathbf{X}_i$ represents the success or failure of the i^{th} trial of your experiment. What is the probability of success, that maximizes the likelihood of the observed samples.

Suppose for $1 \leq i \leq n$ $\mathbf{X}_i = \begin{cases} 1, & \text{if success} \\ 0, & \text{otherwise} \end{cases}$

So, $L = \prod_{i=1}^n \mathbf{P}(\mathbf{X}_i|\theta) = \theta^{\sum_{i=1}^n \mathbf{X}_i} (1 - \theta)^{\sum_{i=1}^n (1 - \mathbf{X}_i)}$ $\log(L) = \sum_{i=1}^n \log(\mathbf{P}(\mathbf{X}_i|\theta)) = \sum_{i=1}^n \mathbf{X}_i \log(\theta) + \sum_{i=1}^n (1 - \mathbf{X}_i) \log(1 - \theta)$

$$\begin{aligned} \frac{d \log(L)}{d\theta} &= \frac{d \sum_{i=1}^n \mathbf{X}_i \log(\theta) + \sum_{i=1}^n (1 - \mathbf{X}_i) \log(1 - \theta)}{d\theta} \\ &= \frac{\sum_{i=1}^n \mathbf{X}_i}{\theta} - \frac{\sum_{i=1}^n (1 - \mathbf{X}_i)}{1 - \theta} \end{aligned}$$

MLE of Binomial Distribution

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are independent random samples from a binomial distribution
What is the probability of success, that maximizes the likelihood of the observed samples.

Suppose for $1 \leq i \leq n$, $0 \leq \mathbf{X}_i \leq n$

$$\mathbf{P}(\mathbf{X}_i|\theta) = \binom{n}{\mathbf{X}_i} \theta^{\mathbf{X}_i} \cdot (1 - \theta)^{n - \mathbf{X}_i} \quad L = \prod_{i=1}^n \binom{n}{\mathbf{X}_i} \theta^{\mathbf{X}_i} \cdot (1 - \theta)^{n - \mathbf{X}_i}$$

$$\log(L) = \sum_{i=1}^n \left(\log \binom{n}{\mathbf{X}_i} + \mathbf{X}_i \log(\theta) + (n - \mathbf{X}_i) \log(1 - \theta) \right)$$

$$\frac{d \log(L)}{d\theta} = \frac{\sum_{i=1}^n \mathbf{X}_i}{\theta} - \frac{\sum_{i=1}^n (n - \mathbf{X}_i)}{1 - \theta}$$

MLE of Geometric Distribution

Let $X_1, X_2, \dots, X_n$ are independent random samples from a geometric distribution
What is the probability of success, that maximizes the likelihood of the observed samples.

Suppose for $1 \leq i \leq n$, $1 \leq X_i \leq \infty$

$$P(X_i|\theta) = (1 - \theta)^{X_i-1}\theta \quad L = \prod_{i=1}^n (1 - \theta)^{X_i-1}\theta$$

$$\log(L) = \sum_{i=1}^n ((X_i - 1) \log(1 - \theta) + \log(\theta))$$

$$\frac{d \log(L)}{d\theta} = \frac{n - \sum_{i=1}^n X_i}{1 - \theta} + \frac{n}{\theta}$$